Black hole shadows in Topology-Generated Potential: photon sphere shift and falsifiable predictions for the Event Horizon Telescope

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Abstract

Topology-Generated Potential (TGP) is a scalar-field framework in which the physical constants c, $\hbar$, and G are field-dependent while the Planck length $\ell_{\rm P}$ remains invariant. The weak-field, static, spherically symmetric metric takes the exponential (Yilmaz–Rosen) form $\psi = \exp(-r_{\rm s}/r)$, which reproduces all post-Newtonian predictions of general relativity (GR) with $\gamma = \beta = 1$.

We present three levels of analysis. **(1) Exponential extrapolation:** if taken to hold at all radii, the exponential metric places the photon sphere at $R_{\rm ph} = r_{\rm s}/2$ in curvature coordinates, yielding a shadow $\sim 48\%$ smaller than Schwarzschild. **(2) Consistency check:** we prove analytically that the exponential metric does *not* satisfy the full nonlinear TGP field equation — the residual is $\mathcal{O}(\beta)$ at all radii — and that in isotropic coordinates the function $r \cdot \psi(r)$ is monotonically increasing, so the exponential metric possesses *no photon sphere* in coordinate-invariant terms. The "$R_{\rm ph} = r_{\rm s}/2$" result is an artifact of applying the curvature-coordinate photon-sphere formula to a metric form that is not the true strong-field solution. **(3) Exact soliton:** we discover that the substitution $u = \psi^3$ linearizes the TGP kinetic operator to the Laplace equation $\nabla^2 u = 0$, yielding the closed-form BH soliton $\psi(r) = (1 + 3r_{\rm s}/r)^{1/3}$. This solution has a frozen interior ($g_{tt} \to 0$, $\psi \to \infty$), no singularity, and crucially **no photon sphere**: the function $h(r) = r\,\psi(r)$ is strictly monotonically increasing. The absence of a photon sphere is a falsifiable prediction of TGP.

Black hole thermodynamics is preserved exactly ($S_{\rm BH} = k_B A/4\ell_{\rm P}^2$) and the no-singularity theorem ($K \to 0$) is robust. Ringdown frequency shifts ($\delta f_0 \approx +0.6\%$, $\delta f_2 \approx -0.2\%$) provide independent gravitational-wave tests.

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1 Introduction

The direct imaging of the supermassive black hole in M87 by the Event Horizon Telescope (EHT) collaboration [?, ?] and the subsequent image of Sagittarius A* [?] have opened an unprecedented window on strong-field gravity. The bright emission ring observed around each compact object is interpreted as the lensed image of accreting plasma, with its diameter set by the *critical impact parameter* $b_{\rm crit}$ — the closest approach at which a photon can still escape to infinity. In general relativity, for a Schwarzschild black hole of mass M,

$$b_{\rm crit}^{\rm GR} = 3\sqrt{3}\,\frac{GM}{c_0^2} \approx 2.598\, r_{\rm s}, \qquad r_{\rm s} \equiv \frac{2GM}{c_0^2}. \tag{1}$$

The EHT measurements of M87* yield a shadow angular diameter of 42 ± 3 μas [?], consistent with the GR prediction to within roughly 10%. For Sgr A*, the measured diameter is 48.7 ± 7 μas [?], again broadly consistent with Schwarzschild.

These observations already constrain a number of modified gravity theories and parametrized deviations from the Kerr metric [?, ?, ?]. At the same time, the uncertainties — dominated by source morphology and interstellar scattering rather than by instrumental noise — leave room for $\mathcal{O}$(10–30%) deviations in the shadow size [?, ?]. Any theory that makes a sharp, quantitative prediction for the shadow diameter is therefore immediately testable.

Topology-Generated Potential (TGP) is a framework in which spacetime itself is an emergent quantity generated by a scalar substrate field Φ [?, ?]. The fundamental constants c, $\hbar$, and G become field-dependent:

$$c(\Phi) = c_0\left(\frac{\Phi_0}{\Phi}\right)^{1/2}, \quad \hbar(\Phi) = \hbar_0\left(\frac{\Phi_0}{\Phi}\right)^{1/2}, \quad G(\Phi) = G_0\,\frac{\Phi_0}{\Phi}, \tag{2}$$

where c_0, $\hbar_0$, G_0 are the vacuum values measured at spatial infinity ($\Phi = \Phi_0$). A key structural result is that the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3}$ is *invariant* under these scalings, ensuring that quantum-gravitational physics remains universal.

In the weak-field regime, TGP reproduces the Schwarzschild solution to all post-Newtonian (PPN) orders with $\gamma = \beta = 1$, matching every Solar-System and binary-pulsar test catalogued by Will [?]. The speed-of-gravity constraint from GW170817/GRB 170817A [?] is likewise satisfied. In the strong-field regime, however, the exponential form of the TGP metric produces qualitative departures: there is no true event horizon, no curvature singularity, and the photon sphere shifts inward.

The purpose of this paper is to work out the observational consequences of the TGP metric for the black hole shadow. We derive the photon sphere radius and critical impact parameter, tabulate predictions for M87* and Sgr A*, discuss the no-singularity theorem and thermodynamic consistency, and assess the current observational status with explicit caveats.

Plan. Section 2 reviews the TGP exponential metric. Section 3 derives the photon orbits. Section 4 presents the no-singularity theorem. Section 5 verifies black hole thermodynamics. Section 6 tabulates observational predictions. Sections 8 and 9 discuss limitations and conclude.

2 TGP exponential metric

2.1 Field equation and static solution

The TGP substrate field $\Phi(\mathbf{x}, t)$ satisfies a nonlinear wave equation on flat background whose detailed derivation is given in [?, ?]. For the purposes of this paper, we need only the static, spherically symmetric vacuum solution.

In the weak-field (Newtonian) limit, $\Phi \approx \Phi_0$ and the gravitational potential is $U = -GM/r$. The TGP line element for a static, spherically symmetric source takes the *exponential metric* form:

$$\boxed{\mathrm{d}s^2 = -\frac{c_0^2}{\psi}\,\mathrm{d}t^2 + \psi\,(\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2),} \tag{3}$$

where

$$\psi(r) \equiv \exp\!\left(\frac{2U(r)}{c_0^2}\right) = \exp\!\left(-\frac{r_{\rm s}}{r}\right), \qquad r_{\rm s} = \frac{2GM}{c_0^2}. \tag{4}$$

Here $\mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2$ is the standard unit-sphere element. The isotropic radial coordinate r is used throughout.

2.2 Key properties

(i) **No coordinate singularity.** Since $\psi = e^{-r_{\rm s}/r} > 0$ for all $r > 0$, the metric coefficients are everywhere finite and positive. There is no radius at which $g_{tt} = 0$ or $g_{rr} = \infty$; in particular, the surface $r = r_{\rm s}$ is *not* a horizon in the Schwarzschild sense. The light cones pinch continuously but never close.

(ii) **Weak-field agreement with GR.** Expanding ψ to any finite order in $r_{\rm s}/r$ and comparing with the PPN-expanded Schwarzschild metric in isotropic coordinates, one finds exact

agreement at every order:

$$g_{tt} = -c_0^2 e^{+r_s/r} = -c_0^2\left(1 + \frac{r_s}{r} + \frac{1}{2}\frac{r_s^2}{r^2} + \cdots\right), \tag{5}$$

$$g_{rr} = e^{-r_s/r} = 1 - \frac{r_s}{r} + \frac{1}{2}\frac{r_s^2}{r^2} - \cdots . \tag{6}$$

In particular, the PPN parameters are $\gamma = \beta = 1$, identical to GR [**?**].

(iii) **Strong-field departure.** The exponential and Schwarzschild metrics agree perturbatively but differ non-perturbatively. The Schwarzschild metric (in isotropic coordinates) has $g_{tt} = 0$ at $r = r_s/4$ (the horizon in isotropic form), while the TGP metric has $g_{tt} \to -c_0^2$ as $r \to 0$ after passing through a minimum. This is the root of the shadow-size discrepancy derived below.

2.3 Relation to the substrate field

The conformal factor ψ is related to the substrate field by $\psi = \Phi/\Phi_0 = \chi$. In the weak-field exterior, $\chi \approx 1 - r_s/(2r) + \cdots$, while in the strong-field interior $\chi \gg 1$ (approaching the "frozen" phase described in Section 4). The dimensionless field χ is the natural variable for discussing both the duality structure and the curvature invariants.

3 Photon orbits in the TGP metric

The photon sphere and critical impact parameter are most cleanly derived in *curvature* (Schwarzschild-like) coordinates, where the angular part of the metric is manifestly $R^2\,\mathrm{d}\Omega^2$ with R the areal radius. We therefore begin by writing the TGP exponential metric in this form.

3.1 Exponential metric in curvature coordinates

The isotropic-coordinate line element (3) is equivalent, after a standard coordinate transformation, to the *Yilmaz–Rosen* form [**?**, **?**]:

$$\mathrm{d}s^2 = -e^{-r_s/R}\,c_0^2\,\mathrm{d}t^2 + e^{+r_s/R}\,\mathrm{d}R^2 + R^2\,\mathrm{d}\Omega^2, \tag{7}$$

where R is the areal radius and $r_s = 2GM/c_0^2$ as before. We define

$$A(R) \equiv e^{-r_s/R}, \qquad B(R) \equiv e^{+r_s/R}. \tag{8}$$

Note that $A(R) > 0$ and $B(R) > 0$ for all $R > 0$: the metric is regular everywhere outside the origin.

For comparison, the Schwarzschild metric has $A_{\mathrm{GR}} = 1 - r_s/R$ and $B_{\mathrm{GR}} = (1 - r_s/R)^{-1}$, with $A_{\mathrm{GR}} = 0$ at the horizon $R = r_s$.

3.2 Null geodesics

We restrict to the equatorial plane $\theta = \pi/2$ (spherical symmetry). The Killing vectors ∂_t and ∂_φ yield conserved energy E and angular momentum L:

$$E = A(R)\,c_0^2\,\dot{t}, \qquad L = R^2\,\dot{\varphi}, \tag{9}$$

where the dot denotes differentiation with respect to an affine parameter. The null condition $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$ gives the radial equation

$$\dot{R}^2 = \frac{E^2}{A(R)\,B(R)} - \frac{L^2}{R^2\,B(R)} = \frac{1}{B}\left(\frac{E^2}{A} - \frac{L^2}{R^2}\right). \tag{10}$$

Since $A\,B = e^{-r_s/R} \cdot e^{+r_s/R} = 1$ for the exponential metric, this simplifies to

$$\dot{R}^2 = E^2\, e^{r_s/R} - \frac{L^2}{R^2}\, e^{r_s/R} = e^{r_s/R}\left(E^2 - \frac{L^2}{R^2}\right). \tag{11}$$

Defining the impact parameter $b \equiv L/E$, the turning point $\dot{R} = 0$ occurs where

$$b^2 = R^2. \tag{12}$$

This would seem to imply that the closest approach is simply $R = b$ with no deflection. However, this is an artifact of the product $AB = 1$ simplifying the radial equation. The correct effective potential analysis retains the full geodesic equation.

Let us instead use the standard photon-sphere criterion for a general static spherically symmetric metric [**?**, **?**].

3.3 Photon sphere condition

For a metric of the form

$$\mathrm{d}s^2 = -A(R)\, c_0^2\, \mathrm{d}t^2 + B(R)\, \mathrm{d}R^2 + R^2\, \mathrm{d}\Omega^2, \tag{13}$$

the effective potential for null geodesics, written in terms of the impact parameter b, takes the form

$$V_{\text{eff}}(R) = \frac{A(R)}{R^2}. \tag{14}$$

The photon sphere is located where V_{eff} has an extremum:

$$\frac{\mathrm{d}}{\mathrm{d}R}\left[\frac{A(R)}{R^2}\right] = 0 \qquad \Longleftrightarrow \qquad \frac{A'(R)}{A(R)} = \frac{2}{R}. \tag{15}$$

GR (Schwarzschild). With $A = 1 - r_s/R$:

$$\frac{A'}{A} = \frac{r_s/R^2}{1 - r_s/R} = \frac{2}{R} \qquad \Longrightarrow \qquad R_{\text{ph}}^{\text{GR}} = \frac{3}{2}\, r_s. \tag{16}$$

TGP (exponential). With $A = e^{-r_s/R}$:

$$A'(R) = \frac{r_s}{R^2}\, e^{-r_s/R}, \qquad \frac{A'}{A} = \frac{r_s}{R^2}. \tag{17}$$

Setting this equal to $2/R$:

$$\frac{r_s}{R^2} = \frac{2}{R} \qquad \Longrightarrow \qquad \boxed{R_{\text{ph}}^{\text{TGP}} = \frac{r_s}{2} = \frac{GM}{c_0^2}.} \tag{18}$$

The photon sphere in TGP lies at one-third the GR value: $R_{\text{ph}}^{\text{TGP}} = R_{\text{ph}}^{\text{GR}}/3$.

3.4 Critical impact parameter

The critical impact parameter is the value of b for a photon on the unstable circular orbit at R_{ph}:

$$b_{\text{crit}} = \frac{R_{\text{ph}}}{\sqrt{A(R_{\text{ph}})}}. \tag{19}$$

GR (Schwarzschild).

$$b_{\text{crit}}^{\text{GR}} = \frac{3r_s/2}{\sqrt{1 - 2/3}} = \frac{3r_s/2}{\sqrt{1/3}} = \frac{3\sqrt{3}}{2}\, r_s = 3\sqrt{3}\, \frac{GM}{c_0^2} \approx 2.598\, r_s. \tag{20}$$

TGP (exponential).

$$b_{\rm crit}^{\rm TGP} = \frac{r_{\rm s}/2}{\sqrt{e^{-r_{\rm s}/(r_{\rm s}/2)}}} = \frac{r_{\rm s}/2}{\sqrt{e^{-2}}} = \frac{r_{\rm s}}{2}\, e = \frac{GM\, e}{c_0^2} \approx 1.359\, r_{\rm s}. \tag{21}$$

The ratio of shadow sizes is therefore

$$\boxed{\frac{b_{\rm crit}^{\rm TGP}}{b_{\rm crit}^{\rm GR}} = \frac{(r_{\rm s}/2)\, e}{(3\sqrt{3}/2)\, r_{\rm s}} = \frac{e}{3\sqrt{3}} \approx \frac{2.718}{5.196} \approx 0.523.} \tag{22}$$

The TGP shadow is approximately **48% smaller** than the Schwarzschild prediction.

Remark 3.1 (Relation to the 37% estimate). A preliminary estimate based on the isotropic-coordinate form of the metric — identifying the photon sphere with the isotropic coordinate value $r_{\rm ph} = r_{\rm s}$ and computing $b_{\rm crit} = r_{\rm s}\sqrt{e}$ — yielded a ratio $\sqrt{e}/(3\sqrt{3}/2) \approx 0.635$, corresponding to a 37% reduction. The discrepancy arises because the isotropic coordinate r does not directly measure circumferential radius; the physically meaningful calculation in curvature coordinates gives $R_{\rm ph} = r_{\rm s}/2$ and $b_{\rm crit} = (r_{\rm s}/2)\, e$, producing the larger 48% reduction. The qualitative conclusion — that TGP predicts a *substantially* smaller shadow — is the same in both analyses.

3.5 Consistency check: does the exponential metric solve the TGP field equation?

The preceding derivation assumed that the exponential metric holds at all radii, including the photon sphere at $R = r_{\rm s}/2$. We now check whether this assumption is self-consistent.

Residual of the field equation. The full TGP field equation in vacuum ($\rho = 0$) is

$$\psi'' + \frac{2}{r}\,\psi' + 2\,\frac{(\psi')^2}{\psi} + \beta\,\psi^2 - \gamma\,\psi^3 = 0, \tag{23}$$

where $\beta = \gamma$ (vacuum condition). For the exponential ansatz $\psi_{\rm exp} = e^{-r_{\rm s}/r}$, the kinetic part evaluates to

$$\psi'' + \frac{2}{r}\psi' + \frac{2(\psi')^2}{\psi} = \frac{3r_{\rm s}^2}{r^4}\, e^{-r_{\rm s}/r}, \tag{24}$$

while the potential part is $\beta e^{-2r_{\rm s}/r}(1 - e^{-r_{\rm s}/r})$. The total residual

$$\mathcal{R}(r) = \frac{3r_{\rm s}^2}{r^4}\, e^{-r_{\rm s}/r} + \beta\, e^{-2r_{\rm s}/r} - \gamma\, e^{-3r_{\rm s}/r} \tag{25}$$

is **strictly positive** for all $r > 0$ when $\beta = \gamma > 0$: both the kinetic term (> 0) and the potential term $\beta e^{-2r_{\rm s}/r}(1 - e^{-r_{\rm s}/r}) > 0$ (since $e^{-r_{\rm s}/r} < 1$) are individually positive.

Proposition 3.2 (Exponential is not the strong-field solution)**.** *The exponential metric $\psi = e^{-r_{\rm s}/r}$ does not satisfy the full nonlinear TGP field equation* (23) *at any radius. It is an approximate solution valid only in the asymptotic regime $r \gg r_{\rm s}$ where $\mathcal{R}(r) \to 0$.*

Absence of photon sphere in isotropic coordinates. The photon sphere condition in isotropic coordinates is $\frac{d}{dr}[r \cdot \psi(r)] = 0$ (Section 3.3). For the exponential metric:

$$\frac{d}{dr}\Big[r\, e^{-r_{\rm s}/r}\Big] = e^{-r_{\rm s}/r}\Big(1 + \frac{r_{\rm s}}{r}\Big) > 0 \quad \text{for all } r > 0. \tag{26}$$

The function $r \cdot \psi$ is **monotonically increasing**: there is no extremum and hence *no photon sphere* in coordinate-invariant terms. The result $R_{\rm ph} = r_{\rm s}/2$ obtained in Section 3.3 arises from applying the curvature-coordinate formula $A'(R)/A(R) = 2/R$ to the Yilmaz–Rosen form, but this form is not the actual strong-field TGP solution (Proposition 3.2).

Remark 3.3 (Status of the 48% result). The 48% shadow reduction derived in Section 3.4 is a **conditional result**: it holds *if and only if* the exponential metric is valid at $R = r_{\rm s}/2$. Proposition 3.2 shows this assumption is inconsistent. The actual TGP shadow size requires the full nonlinear profile $\psi(r)$, derived in Section 7.

3.6 Summary table: GR vs TGP

Table 1: Key black hole observables in GR (Schwarzschild) and TGP (exponential extrapolation, conditional — see Proposition 3.2). Areal radius R throughout.

Observable	Schwarzschild (GR)	TGP (exponential)
Horizon radius R_H	$r_\mathrm{s} = 2GM/c_0^2$	None (no horizon)
Photon sphere R_ph	$\frac{3}{2} r_\mathrm{s}$	$\frac{1}{2} r_\mathrm{s}$
Critical impact parameter b_crit	$\frac{3\sqrt{3}}{2}\, r_\mathrm{s} \approx 2.598\, r_\mathrm{s}$	$\frac{e}{2}\, r_\mathrm{s} \approx 1.359\, r_\mathrm{s}$
Shadow ratio $b_\mathrm{crit}^\mathrm{TGP}/b_\mathrm{crit}^\mathrm{GR}$	1	$e/(3\sqrt{3}) \approx 0.523$
Curvature singularity	Yes ($R = 0$)	No ($K \to 0$)
Event horizon	Yes	No (causal freezing)
PPN parameters γ, β	$1, 1$	$1, 1$

4 No-singularity theorem

4.1 Three mechanisms of singularity avoidance

In TGP, the field-dependent constants provide three independent mechanisms that prevent the formation of a curvature singularity:

(1) **Self-regulated gravity:** $G(\Phi) = G_0\, \Phi_0/\Phi \to 0$ as $\Phi \to \infty$. The gravitational coupling itself vanishes in the strong-field interior, preventing further collapse.

(2) **Causal freezing:** $c(\Phi) = c_0 \sqrt{\Phi_0/\Phi} \to 0$ as $\Phi \to \infty$. Light cones close continuously; no signal can propagate into the deep interior. The region is *causally frozen* — operationally indistinguishable from a horizon, but without the global causal structure of a true event horizon.

(3) **Quantum suppression:** $\hbar(\Phi) = \hbar_0 \sqrt{\Phi_0/\Phi} \to 0$ as $\Phi \to \infty$. Quantum fluctuations, which in standard GR are expected to become important near a singularity, are *suppressed* rather than enhanced.

The combination of these three effects produces a "frozen star" interior: matter does not collapse to a point but is trapped in a region of extreme field density where all dynamical processes cease.

4.2 Kretschner scalar

For the exponential metric (7), the Kretschner scalar (the square of the Riemann tensor) can be computed directly. Introducing the dimensionless variable $\chi \equiv \Phi/\Phi_0$ and comparing with the Schwarzschild result K_GR at the same coordinate radius:

$$K_\mathrm{TGP}(\chi) = \frac{K_\mathrm{GR}}{\chi^4}. \tag{27}$$

As $\chi \to \infty$ (the strong-field interior), $K_\mathrm{TGP} \to 0$: curvature *decreases* rather than diverging. This is the opposite of the GR behavior and is a direct consequence of the exponential metric structure.

Proposition 4.1 (No-singularity theorem)**.** *In the TGP exponential metric, all polynomial curvature invariants remain bounded for $r > 0$. In particular, $K_\mathrm{TGP}(r) \to 0$ as $r \to 0$.*

Proof. The Kretschner scalar for the metric (7) with $\phi = -r_\mathrm{s}/(2R)$ is a sum of terms proportional to $e^{-nr_\mathrm{s}/R} \cdot (r_\mathrm{s}/R)^k$ for various positive n and k. As $R \to 0$, each such term vanishes because $e^{-nr_\mathrm{s}/R}$ decays faster than any power of R. Hence $K \to 0$. □

4.3 Duality $\chi \leftrightarrow 1/\chi$

The TGP framework possesses an *operational duality* under the inversion $\chi \mapsto 1/\chi$ [?]. Under this map:

- The cosmological vacuum ($\chi \to 0$, absence of spacetime) maps to the black hole interior ($\chi \to \infty$, frozen spacetime).
- In both limits, the information capacity $\mathcal{I} = 0$: no signal can propagate.
- The Planck length is invariant: $\ell_{\mathrm{P}} \mapsto \ell_{\mathrm{P}}$.

This duality is *operational* (informational), not a full symmetry of the field equations. Both boundaries of the existence sector $\mathcal{S}_1$ — the "nothingness from deficit" ($\chi = 0$) and the "nothingness from excess" ($\chi \to \infty$) — are functionally equivalent from the standpoint of an external observer.

The physical picture is that of a **frozen star**: the interior is not empty (no singularity) but is a region of extremely dense substrate field in which all physical processes have ceased. This is the TGP analog of the "gravastar" or "frozen star" proposals in the literature, but it arises naturally from the field-dependent constants rather than being imposed by hand.

5 Black hole thermodynamics in TGP

A crucial consistency check for any modified gravity theory is whether black hole thermodynamics survives. In TGP, the Bekenstein–Hawking entropy formula holds *exactly*, with a remarkable cancellation rooted in the Planck-length invariance.

5.1 Bekenstein–Hawking entropy

The entropy of a compact object with horizon-area A and local field value Φ_H on its surface is

$$S_{\mathrm{BH}} = \frac{k_B\, c(\Phi_H)^3}{4\, G(\Phi_H)\, \hbar(\Phi_H)}\, A. \tag{28}$$

Substituting the TGP scalings (2):

$$\frac{c^3}{G\,\hbar} = \frac{c_0^3 (\Phi_0/\Phi_H)^{3/2}}{G_0(\Phi_0/\Phi_H)\cdot \hbar_0(\Phi_0/\Phi_H)^{1/2}} = \frac{c_0^3}{G_0\,\hbar_0}\,\frac{(\Phi_0/\Phi_H)^{3/2}}{(\Phi_0/\Phi_H)^{3/2}} = \frac{c_0^3}{G_0\,\hbar_0}. \tag{29}$$

The exponents cancel exactly: $\frac{3}{2} = 1 + \frac{1}{2}$. This is a direct consequence of $\ell_{\mathrm{P}} = \sqrt{\hbar G/c^3} = \mathrm{const}$. Therefore:

$$\boxed{S_{\mathrm{BH}} = \frac{k_B\, A}{4\,\ell_{\mathrm{P}}^2},} \tag{30}$$

which is the standard Bekenstein–Hawking formula expressed in vacuum constants — **completely independent of the local field Φ_H** [?, ?].

5.2 Hawking temperature

The Hawking temperature in TGP is modified by the local value of $\hbar$ at the horizon [?]:

$$T_H^{\mathrm{TGP}} = T_H^{\mathrm{GR}} \cdot \sqrt{\frac{\Phi_0}{\Phi_H}} \ll T_H^{\mathrm{GR}}, \tag{31}$$

since $\Phi_H \gg \Phi_0$ near the horizon. The suppression of Hawking radiation is consistent with the frozen-star picture: quantum effects ($\hbar \to 0$) are suppressed where the field is strongest.

5.3 First law of thermodynamics

The first law $\mathrm{d}E = T_H\,\mathrm{d}S_{\mathrm{BH}}$ is preserved. Since S_{BH} is Φ-independent (Eq. 30) and depends only on the ADM mass through $A = 16\pi G_0^2 M^2/c_0^4$ (at fixed Φ_H), differentiation with respect to M reproduces the standard thermodynamic identity [?].

Remark 5.1 (Consistency with Wald entropy). The Wald entropy formula [?] for diffeomorphism-invariant theories gives $S = -2\pi \oint (\partial\mathcal{L}/\partial R_{\mu\nu\rho\sigma})\,\hat{\epsilon}_{\mu\nu}\hat{\epsilon}_{\rho\sigma}\,\mathrm{d}\Sigma$. In TGP, the effective gravitational Lagrangian inherits the Einstein–Hilbert form with field-dependent G; the Wald calculation then reproduces (30) because the Φ-dependence enters only through $G(\Phi)$ and cancels in the entropy formula by the same mechanism as (29).

6 Observational predictions

6.1 Shadow sizes for M87* and Sgr A*

Table 2 compiles the predicted shadow angular diameters for the two EHT targets. The GR predictions use the Schwarzschild metric (spin effects are subdominant for the shadow size at current precision [?]).

Table 2: Shadow angular diameters for M87* and Sgr A* in GR and TGP (exponential extrapolation). The TGP values are **conditional**: they assume the exponential metric holds at $R = r_{\mathrm{s}}/2$, which Proposition 3.2 shows is inconsistent. The true TGP prediction requires the full nonlinear profile (Section 7). Observed values from [?, ?].

Source	GR prediction	TGP prediction	Observed
M87*	$\sim 42\ \mu$as	$\sim 22\ \mu$as	$42 \pm 3\ \mu$as
Sgr A*	$\sim 53\ \mu$as	$\sim 28\ \mu$as	$48.7 \pm 7\ \mu$as

6.2 Tension with EHT data

The TGP prediction of a $\sim 48\%$ smaller shadow is in **significant tension** with both EHT measurements. For M87*, the predicted $\sim 22\ \mu$as lies well outside the 1σ error bar of $42 \pm 3\ \mu$as. For Sgr A*, the predicted $\sim 28\ \mu$as is likewise incompatible with the observed $48.7 \pm 7\ \mu$as.

6.3 Critical caveat: strong-field extrapolation

The 48% **figure rests on the assumption that the exponential metric** (7) **holds unchanged all the way down to the photon sphere at** $R = r_{\mathrm{s}}/2$**.** This is a strong assumption. In TGP, the exponential form is derived as the unique conformal metric consistent with the field equation in the *weak-to-moderate* field regime. The full radial profile $\psi(r)$ in the strong-field regime — where the nonlinear self-interaction of Φ becomes important — is an **open problem**.

Several factors could moderate the discrepancy:

(a) **Nonlinear corrections.** The TGP field equation contains cubic and quartic self-interaction terms $(\beta\chi^2 - \gamma\chi^3)$ that are negligible at large r but could modify $\psi(r)$ significantly for $r \lesssim r_{\mathrm{s}}$.

(b) **Interior structure.** Unlike Schwarzschild, the TGP compact object has a finite interior (the frozen core). The transition from the exterior exponential metric to the interior could shift the effective photon sphere outward.

(c) **Accretion morphology.** The EHT "shadow" is not a direct measurement of b_{crit} but of the bright emission ring, which depends on the accretion flow geometry. In a modified metric, the ring-to-shadow mapping could differ.

We therefore present the 48% **reduction as an upper bound on the deviation**, obtained from the simplest extrapolation of the TGP metric. The true deviation, once the full strong-field solution is known, is expected to be smaller. Even a 10–20% deviation would be a powerful discriminator between TGP and GR with next-generation EHT data [**?**].

6.4 Gravitational-wave predictions

Independent of the shadow measurement, TGP predicts modifications to the ringdown spectrum of merging compact objects:

- **Breathing mode** ($\ell = 0$): frequency shift $\delta f_0 \approx +0.6\%$ relative to GR [**?**].
- **Quadrupole mode** ($\ell = 2$): frequency shift $\delta f_2 \approx -0.2\%$.

These shifts are below the sensitivity of current LIGO/Virgo detectors but are within reach of third-generation instruments (Einstein Telescope, Cosmic Explorer) and of LISA for supermassive binary mergers [**?**, **?**].

Additionally, the scalar nature of the TGP field predicts a *breathing polarization* in the stochastic gravitational-wave background, with a characteristic strain $h_{\rm br} \sim 10^{-17}$ at nanohertz frequencies — potentially detectable by the Square Kilometre Array (SKA) through pulsar timing arrays [**?**].

7 Strong-field solution

The consistency analysis of Section 3.5 demonstrates that the exponential metric is insufficient in the strong-field regime. We now derive the *exact* strong-field solution and determine its implications for the shadow.

7.1 The vacuum BH soliton

The TGP vacuum field equation in static spherical symmetry is

$$\psi'' + \frac{2}{r}\psi' + \frac{2(\psi')^2}{\psi} + \beta\psi^2(1-\psi) = 0, \tag{32}$$

where the vacuum condition $\beta = \gamma\Phi_0$ has been used and $\psi = \Phi/\Phi_0$.

The kinetic operator admits a remarkable simplification via the substitution $u \equiv \psi^3$:

$$\psi'' + \frac{2}{r}\psi' + \frac{2(\psi')^2}{\psi} = \frac{1}{3\psi^2}\left(u'' + \frac{2}{r}u'\right) = \frac{1}{3\psi^2}\nabla^2 u, \tag{33}$$

where ∇^2 is the flat spherical Laplacian.

7.1.1 Physical value of β

The scalar substrate mass is $m_{\rm sp} \sim H_0$ (Section **??**), giving a dimensionless $\beta \sim (r_{\rm s} H_0/c_0)^2$. For a stellar-mass black hole ($r_{\rm s} \sim 10^4\,{\rm m}$), $\beta \sim 10^{-44}$; for M87* ($r_{\rm s} \sim 10^{13}\,{\rm m}$), $\beta \sim 10^{-26}$. At the scale of the photon sphere ($r \sim r_{\rm s}$), the self-interaction is negligible: $\beta\psi^2(1-\psi) \ll |\nabla^2\psi/\psi|$ by a factor $\sim 10^{-26}$ or smaller.

7.1.2 Exact solution for $\beta = 0$

Setting $\beta = 0$ in (32) and using (33):

$$\nabla^2 u = u'' + \frac{2}{r}u' = 0. \tag{34}$$

This is simply the Laplace equation, with general solution $u = C_1 + C_2/r$. Imposing $\psi(\infty) = 1$ gives $C_1 = 1$; $\psi(0) \to \infty$ requires $C_2 > 0$. Thus:

$$\boxed{\psi(r) = \left(1 + \frac{C}{r}\right)^{1/3}, \qquad C = 3\,r_{\rm s},} \tag{35}$$

where the constant $C = 3r_{\rm s}$ is fixed by matching the weak-field PPN expansion $\psi \approx 1 + r_{\rm s}/r$ (using $(1 + 3r_{\rm s}/r)^{1/3} \approx 1 + r_{\rm s}/r + \mathcal{O}(r_{\rm s}^2/r^2)$).

Remark 7.1 (PPN agreement). The soliton metric $g_{rr} = \psi = (1 + 3r_{\rm s}/r)^{1/3} \approx 1 + r_{\rm s}/r$, $g_{tt} = -c_0^2/\psi \approx -c_0^2(1 - r_{\rm s}/r)$, reproduces the PPN parameters $\gamma = \beta_{\rm PPN} = 1$ at first post-Newtonian order, identical to GR.

7.1.3 Properties of the soliton

The soliton (35) has the following properties:

(i) **Frozen interior:** $\psi(r) \to (3r_{\rm s}/r)^{1/3} \to \infty$ as $r \to 0$, giving $g_{tt} = -c_0^2/\psi \to 0$ (time freezes) and $g_{rr} = \psi \to \infty$ (space stretches).

(ii) **No singularity:** $\psi(r)$ is finite and smooth for all $r > 0$. The Kretschner scalar $K \to 0$ at the center.

(iii) **No horizon:** $g_{tt} = -c_0^2\,(1 + 3r_{\rm s}/r)^{-1/3}$ is strictly negative for all $r > 0$; there is no surface where $g_{tt} = 0$.

(iv) **Weak divergence:** $\psi \sim r^{-1/3}$ at the center, so $h(r) = r \cdot \psi \sim r^{2/3} \to 0$ as $r \to 0$.

7.2 Absence of a photon sphere

Proposition 7.2 (No photon sphere)**.** *The TGP soliton $\psi(r) = (1 + 3r_{\rm s}/r)^{1/3}$ has no photon sphere. The function $h(r) = r \cdot \psi(r)$ is strictly monotonically increasing for all $r > 0$.*

Proof. Write $h(r) = (r^3 + 3r_{\rm s}\,r^2)^{1/3}$ and compute

$$h'(r) = \frac{r(3r + 6r_{\rm s})}{3(r^3 + 3r_{\rm s}r^2)^{2/3}} = \frac{r + 2r_{\rm s}}{(r^3 + 3r_{\rm s}r^2)^{2/3}} > 0 \quad \text{for all } r > 0. \tag{36}$$

Since $h'(r) > 0$ everywhere, h has no critical points: no photon sphere exists. □

Remark 7.3 (Failure of the topological argument). An earlier version of this paper claimed that the frozen interior ($\psi \to \infty$) guarantees a photon sphere by the intermediate value theorem. This argument assumed that $h(0^+) \to \infty$ (requiring ψ to diverge *faster* than $1/r$). The actual soliton diverges as $r^{-1/3}$, giving $h(0^+) \to 0$. The photon-sphere existence requires ψ to diverge at least as fast as $1/r$; the TGP kinetic operator $(1/(3\psi^2))\nabla^2(\psi^3)$ prevents this.

7.2.1 Verification: effective potential

The null-geodesic effective potential is $V_{\rm eff}(r) = 1/[\psi(r)^2\,r^2] = (1 + 3r_{\rm s}/r)^{-2/3}/r^2$. One can verify that $dV_{\rm eff}/dr < 0$ for all $r > 0$: $V_{\rm eff}$ is monotonically decreasing with no local maximum, confirming the absence of unstable circular photon orbits.

7.3 Self-consistent correction

The TGP action $S = \int d^4x \sqrt{-g_{\text{eff}}}\,\mathcal{L}$ includes the volume element $\sqrt{-g_{\text{eff}}} = c_0\psi$, which modifies the effective potential via the 4D covariant d'Alembertian. For a static field:

$$\sqrt{-g}\,g^{ij} = c_0\,\delta^{ij} \quad \Rightarrow \quad \Box_g\,\Phi = \frac{1}{\psi}\,\nabla^2_{\text{flat}}\,\Phi. \tag{37}$$

The self-consistent equation replaces the flat potential $\beta\psi^2(1-\psi)$ with $\beta\psi^3(1-\psi)$:

$$\psi'' + \frac{2}{r}\psi' + \frac{2(\psi')^2}{\psi} + \beta\psi^3(1-\psi) = 0. \tag{38}$$

For $\beta = 0$ this is identical to the flat equation, so the soliton (35) is unchanged. For $\beta > 0$, the near-center power law changes from $\psi \sim r^{-1}$ (flat) to $\psi \sim r^{-2/3}$ (self-consistent), giving $h \sim r^{1/3} \to 0$ — an even *weaker* divergence, reinforcing the absence of a photon sphere.

7.4 Robustness against lattice discretization

TGP derives its field equation from a discrete substrate $\Gamma = (V, E)$ via coarse-graining [**?**]. One may ask whether the no-photon-sphere result survives when the continuum Laplacian is replaced by its discrete counterpart on a cubic lattice with block spacing L_B.

7.4.1 Perturbative protection (harmonicity)

The discrete Laplacian admits the Taylor expansion

$$\nabla^2_{\text{disc}}\,u = \nabla^2 u + \frac{L_B^2}{12}\,\nabla^4 u + \mathcal{O}(L_B^4). \tag{39}$$

Since the soliton variable $u = \psi^3$ satisfies $\nabla^2 u = 0$, the bi-Laplacian $\nabla^4 u = \nabla^2(\nabla^2 u) = 0$, and similarly for all higher orders $\nabla^{2n} u = 0$, $n \geq 1$. Consequently:

Proposition 7.4 (Harmonicity protection)**.** *All perturbative lattice corrections to the soliton equation vanish identically. The continuum soliton $\psi = (1 + 3r_{\text{s}}/r)^{1/3}$ is an exact solution of the Taylor-expanded discrete field equation to* ***all orders*** *in L_B.*

7.4.2 Non-perturbative regime: discrete Green's function

At distances $r \lesssim L_B$, the Taylor expansion breaks down. The true discrete equation is $\nabla^2_{\text{disc}}\,u = 0$, whose solution is the lattice Green's function G_{disc}. Unlike the continuum $1/(4\pi r)$, the lattice Green's function is *finite* at the origin: $G_{\text{disc}}(0) = W_3/(6 \cdot 2\pi^2) \approx 0.2527/L_B$ (Watson's triple integral).

This regularizes the soliton: $\psi(0)$ becomes finite rather than divergent. Crucially, this makes the photon-sphere condition *harder* to satisfy, not easier: near the center $\psi' \approx 0$, so $h'(r) = \psi + r\psi' \approx \psi > 0$.

7.4.3 Structural argument: the 1/3 exponent

The absence of a photon sphere rests on a single structural fact. The photon-sphere condition $h'(r_*) = 0$ requires $\psi'/\psi = -1/r$. Since $\psi = u^{1/3}$:

$$\frac{\psi'}{\psi} = \frac{1}{3}\,\frac{u'}{u}. \tag{40}$$

For any positive solution of $\nabla^2 u = 0$ (or its discrete analogue) with $u \to 1 + C/r$, the steepest possible logarithmic gradient is $|u'/u| \to 1/r$ as $r \to 0$. Therefore $|\psi'/\psi| \leq 1/(3r) < 1/r$, and $h' > 0$ everywhere.

The factor $1/3 = 1/(\alpha+1)$ with $\alpha = 2$ (the TGP kinetic exponent) is the *fundamental* reason for the absence of a photon sphere. It is independent of the lattice spacing, lattice geometry, or r_s/L_B ratio.

Remark 7.5 (Critical exponent). A photon sphere would require $p = 1/(\alpha+1) \geq 1$, i.e. $\alpha \leq 0$. The TGP value $\alpha = 2$ is far from this threshold. The no-photon-sphere prediction is therefore robust against any reasonable perturbation of the theory.

7.5 Implications

(i) **No classical shadow:** The TGP soliton has no photon sphere and therefore no classical BH shadow in the GR sense. This result holds for the exact continuum solution, the self-consistent correction, and the discrete-substrate formulation.

(ii) **Observational consequences:** The absence of a photon sphere does *not* mean the object is invisible. It implies (a) no sharp "photon ring" as in GR, (b) a softer brightness gradient at the edge of the dark region, (c) different higher-order photon ring signatures. Current EHT resolution ($\sim 20\,\mu$as) cannot yet distinguish these from the GR prediction with high confidence.

(iii) **Kill-shot test (strengthened):** If future EHT observations (ngEHT) detect sharp photon-ring substructure consistent with a GR photon sphere at $R_{\rm ph} = 3GM/c^2$, this would constitute a **falsification of TGP**. The loophole "unless discrete substrate effects modify the near-center behavior" is now *closed*: the 1/3 exponent structurally forbids a photon sphere for *any* lattice geometry (Proposition **??**, equation (**??**)).

8 Discussion

8.1 Honest assessment

The analysis in this paper has proceeded through three levels of increasing rigor.

Level 1 (exponential extrapolation): the weak-field metric $\psi = e^{-r_s/r}$ gives 48% smaller shadow if taken literally. **Level 2** (consistency check): this metric does not satisfy the TGP field equation and has no photon sphere in coordinate-invariant terms (Proposition 3.2). **Level 3** (exact soliton): the $u = \psi^3$ substitution linearizes the kinetic operator, yielding the exact BH soliton $\psi = (1 + 3r_s/r)^{1/3}$ (equation (35)). This solution has **no photon sphere** (Proposition 7.2).

The no-photon-sphere result is a sharp, falsifiable prediction of TGP. It arises because the nonlinear kinetic operator $(1/(3\psi^2))\nabla^2(\psi^3)$ constrains the soliton to diverge only as $r^{-1/3}$ at the center — too slowly to create the h-function minimum required for a photon sphere.

8.2 Comparison with the topological argument

An earlier version of this work (Proposition 6.1, now corrected) claimed that any ψ with $\psi(0) = \infty$ and $\psi(\infty) = 1$ must have a photon sphere. The error lay in assuming $h(0^+) = r \cdot \psi \to \infty$, which requires ψ to diverge faster than $1/r$. The TGP soliton diverges as $r^{-1/3}$, giving $h(0^+) \to 0$ and h monotonically increasing. This demonstrates that the existence of a photon sphere depends not only on the boundary conditions but on the *divergence rate*, which is determined by the kinetic structure of the field equation.

8.3 Comparison with other modified gravity predictions

Several modified gravity theories predict shadow sizes that deviate from GR [**?**, **?**, **?**]:

- Scalar-tensor theories (Brans–Dicke): typically $\mathcal{O}(1\%)$ deviations for $\omega_{\rm BD} > 40000$.
- Einstein–dilaton–Gauss–Bonnet: up to $\sim 10\%$ for maximal coupling.

- Regular black holes (Bardeen, Hayward): $\sim$ 5–15% deviations depending on the regularization scale.
- Parametrized post-Einsteinian frameworks [**?**, **?**]: continuous deviation parameters.

The TGP prediction (even if moderated to $\sim 20\%$) would be among the *largest* shadow deviations predicted by any viable theory, making it exceptionally testable.

8.4 Distinguishing TGP from GR observationally

Beyond the shadow size, TGP offers several independent observational handles:

(i) **Shadow shape:** The exponential metric predicts a different radial profile of deflection, leading to subtle differences in the photon ring structure (Lyapunov exponents, photon ring autocorrelation).

(ii) **Ringdown spectrum:** The breathing mode shift $\delta f_0 \approx +0.6\%$ provides a gravitational-wave test independent of electromagnetic observations.

(iii) **Gravitational lensing:** Strong lensing near the compact object will show deviations in the Einstein ring radius and time delays.

(iv) **Absence of information paradox:** The frozen-star interior resolves the information paradox: information is not lost but trapped in a region of extremely slow (effectively zero) dynamics. Late-time deviations in Hawking radiation spectrum could provide a test, though this is beyond current observational reach.

9 Conclusion

We have analyzed the black hole shadow problem in TGP at three levels of rigor:

(1) **Exponential extrapolation** (conditional): the weak-field metric $\psi = e^{-r_s/r}$, if taken literally, gives a photon sphere at $R_{\rm ph} = r_s/2$ and a shadow 48% smaller than Schwarzschild. This metric does not satisfy the TGP field equation (Proposition 3.2).

(2) **Exact soliton solution** (new result): the substitution $u = \psi^3$ linearizes the kinetic operator to $\nabla^2 u = 0$, yielding the closed-form BH soliton $\psi = (1+3r_s/r)^{1/3}$. This solution has a frozen interior ($g_{tt} \to 0$), no singularity, no horizon, and **no photon sphere** (Proposition 7.2).

(3) **Robust results:** (a) TGP produces no curvature singularity: $K \to 0$; (b) black hole thermodynamics is preserved: $S_{\rm BH} = k_B A/(4\ell_{\rm P}^2)$; (c) the absence of a photon sphere is a **falsifiable prediction**: detection of sharp photon-ring substructure by ngEHT would rule out TGP in its smooth-field formulation.

The no-photon-sphere result is a consequence of the nonlinear kinetic operator $D_{\rm kin}[\psi] = (1/(3\psi^2))\nabla^2(\psi^3)$, which constrains the soliton divergence to $\psi \sim r^{-1/3}$ — too slow for $h(r) = r\psi(r)$ to develop a minimum. This structural prediction distinguishes TGP from GR in the strong-field regime while remaining fully consistent in the weak field ($\gamma = \beta_{\rm PPN} = 1$).

TGP occupies a distinctive position: *indistinguishable* from GR in every weak-field and PPN test, yet predicting qualitatively different strong-field phenomenology — a frozen-star interior with no photon sphere, testable by next-generation EHT observations.

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